

Empirical formula
percentage oxygen = 53.33 (1)

C : H : O

$\frac{40.0}{12} : \frac{6.67}{1.01} : \frac{53.33}{16} \Rightarrow 1 : 2 : 1 \Rightarrow \text{CH}_2\text{O} \quad (1)$

Mass spectrum
 $M_r = 90 \quad (1)$
molecular formula $\text{C}_3\text{H}_6\text{O}_3 \quad (1)$

m/z of a fragment linked to identity of fragment (1)
e.g. CH_3^+ at 15, $\text{COOH}^+ / \text{CH}_3\text{CH}(\text{OH})^+$ at 45, OH^+ at 17

IR spectrum
peak in range 1650 to 1750 due to $\text{C}=\text{O} \quad (1)$
peak in range 2500 to 3200 / 3200 to 3550 due to $\text{O}-\text{H} \quad (1)$

^1H NMR
4 hydrogen environments / ratio of 1 : 1 : 1 : 3 (1)
any δ value linked to group (1)
e.g. 1 to 1.5 due to CH_3 , 2.5 to 3.0 due to CH_3CH ,
4.0 due to RCOH , 12.5 due to $\text{C}=\text{OOH}$

Identification
X is $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ / 2-hydroxypropanoic acid
do **not** accept 3-hydroxypropanoic acid